

Real-Time FPGA Implementation and Verification of OSA-Aided SBC-BCH FEC for 100G Coherent Optical Access Network

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Abstract We present the first real-time FPGA implementation of an OSA-aided SBC-BCH FEC for 100G coherent optical networks. Featuring a $7.6\mu\text{s}$ round-trip latency, it achieves a 2.5×10^{-10} post-FEC BER at 8.6×10^{-3} pre-FEC BER, confirming stable operation with a PTBC-based coherent DSP.

Introduction

The advent of 6G optical access networks necessitates high-capacity transmission beyond 100Gbps per wavelength. While soft-decision forward error correction (SD-FEC) with low-density parity-check (LDPC) codes provides near-capacity performance for long-haul transmission [1], it requires substantial power consumption and latency, making it unsuitable for cost-sensitive and latency-critical edge and access networks. Consequently, high-gain hard-decision FEC (HD-FEC) with low overhead (typically 10~15%) and low complexity has emerged as a crucial technology for 100G coherent optical access systems [2][3].

Symmetric Block-wise Concatenated BCH (SBC-BCH) codes have been proven to deliver exceptional error-floor performance and coding gain, surpassing conventional product codes by effectively avoiding dominant error-patterns through block-wise data sharing [4]. Recently, an advanced decoding algorithm, namely the one-step-ahead (OSA) aided iterative hard-decision decoding (IHDD), was proposed to further extend the error-correcting capability of BCH codes beyond the conventional t-error limit (from t to t+1), enabling recovery from decoding failures where IHDD stalls, while maintaining performance

competitive with SD-LDPC codes [5]. However, integrating this sophisticated algebraic algorithm into a real-time hardware architecture capable of sustaining 100Gbps throughput has remained a significant engineering challenge.

In this paper, we propose the first hardware architecture and FPGA implementation of the OSA-aided SBC-BCH FEC designed for 100G coherent optical access modules. By employing a highly efficient time-multiplexed parallel processing architecture, we successfully mapped the complex OSA discrepancy calculations and iterative updates into a hardware logic. The real-time performance is extensively verified on a Xilinx Virtex UltraScale+ FPGA [6], successfully demonstrating integration with a 100GE physical coding sublayer (PCS) and a PTBC (Polarization Time Block Code) coherent DSP (CL-iDSP) [7].

The Proposed FEC Hardware Architecture

The proposed FEC architecture operates on an SBC-BCH code utilizing 152 constituent BCH(1023, 953, 7) codes, with an overhead of 15%. As illustrated in Fig. 1, the overall hardware consists of a low latency encoder and a high-performance OSA-aided iterative decoder.

A. Encoder structure

After rate-matching the 100GbE PCS layer

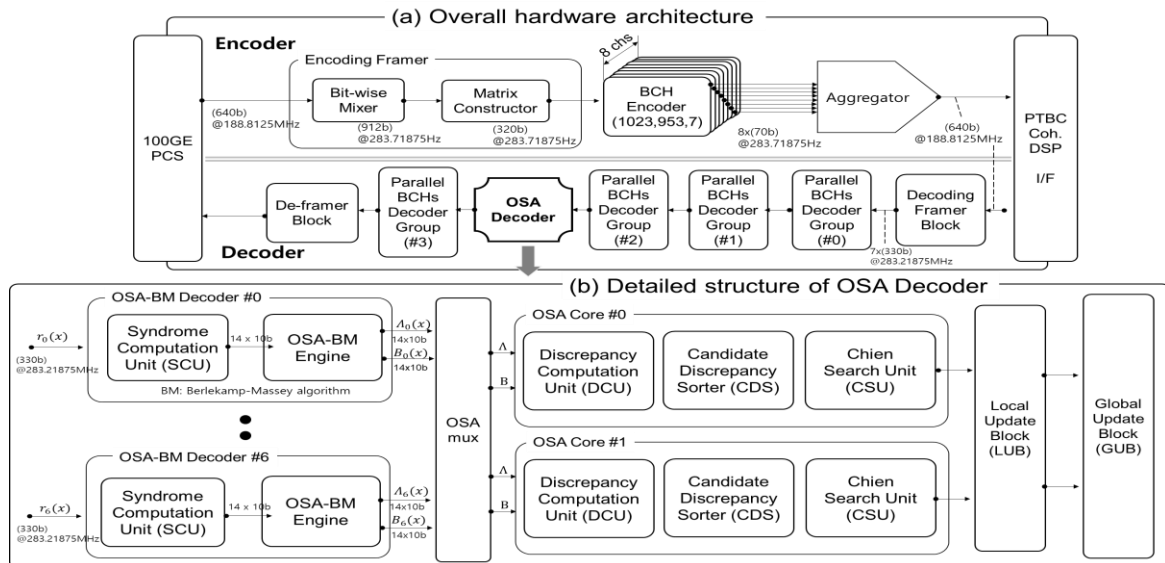


Fig. 1: OSA aided SBC-BCH hardware architecture (a) overall architecture (b) detailed structure of OSA decoder.

output, the encoder receives a 640-bit parallel data stream at 188.81 MHz. The incoming data is processed by the Encoding Framer, which utilizes a bit-wise mixer and a matrix constructor to map the data into the two-dimensional symmetric block-wise structure. The framed data is then distributed across 8 parallel BCH(1023, 953, 7) encoding pipelines operating at 283.2 MHz. Finally, an Aggregator interleaves and multiplexes the systematic information and parity bits into a continuous stream (640-bit at 188.81 MHz) to interface with the PTBC coherent DSP.

B. Decoder: Framer and De-framer Block

On the receiver side, the Decoding Framer converts 640-bit continuous data streams from the PTBC coherent DSP interface into the required 152×152 intersection matrix. This process heavily relies on Block RAMs (BRAM) to temporarily store and reorder the interleaved bits to resolve decoding ambiguity dynamically. The processed parallel data is transferred to the decoder group at a clock speed of 283.2 MHz. After the iterative decoding is completed, the De-framer removes the shared bits among the 152 channels and restores the corrected payload back to the 100GbE standard format.

C. Decoder: Parallel BCH Decoder Group

Instead of instantiating 152 independent BCH decoders which would exceed typical FPGA capacities, we adopt a time-multiplexed parallel architecture. The hardware deploys 7 parallel BCH decoder pipelines. The 152 channels are divided into these 7 parallel groups, sequentially processed over 250 clock cycles to complete a single global iteration. Each BCH decoder pipeline consists of a Syndrome Calculator, a Berlekamp-Massey (BM) Decoder, a Chien Search block, and a Local/Global Update unit. Specifically, the Global-Update Block ensures that when a sub-block is successfully corrected in one channel, the correction is immediately reflected in the shared intersecting channel, significantly accelerating convergence.

D. Decoder: OSA Decoder Core

The OSA decoder acts as a dedicated post-processor to resolve decoding failures when the standard IHDD stalls [5]. As illustrated in Fig. 1(b), the OSA decoding process begins with the OSA-BM Decoder, which consists of a Syndrome Computation Unit (SCU) and an OSA-BM Engine.

To minimize computational latency and complexity, a memory buffer is utilized to store the error-locator polynomial $\Lambda^{(2t)}(x)$ and the scratchpad polynomial $B^{(2t)}(x)$ of the failed constituent codes from the last IHDD iteration. The buffered polynomials are routed through an OSA multiplexer into the OSA Core, which primarily comprises a Discrepancy Computation Unit (DCU), a Candidate Discrepancy Sorter (CDS), and a Chien Search Unit (CSU). Instead of exhaustive searching, the

DCU selectively evaluates the rational discrepancy function, $\tilde{\Delta}_i = \Lambda^{(2t)}(\alpha^{-i}) / (\alpha^{-i} B^{(2t)}(\alpha^{-i}))$, exclusively for bit positions i within the uncorrected blocks identified by the IHDD. The CDS then sorts and accumulates these discrepancy values into a group counter, seeking a unique $\tilde{\Delta}_i$ that appears exactly $t+1$ times (i.e., 8 times for $t=7$). This identified value serves as the valid final discrepancy to update the polynomial to $\Lambda^{(2t)}(x)$. Finally, the CSU pinpoints the extra errors, and the corrections are propagated through the Local and Global Update Blocks (LUB and GUB) to revive the stalled iterative decoding process.

Implementation Results and System Verification

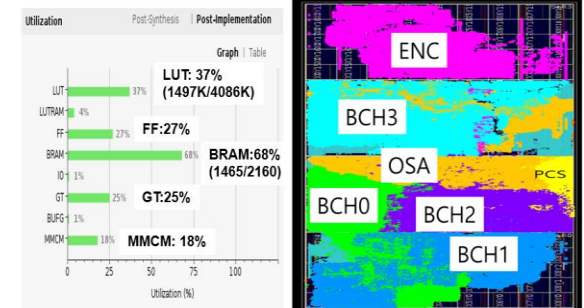
A. FPGA Implementation Results

The proposed FEC algorithm was implemented on a ProFPGA platform featuring a Xilinx Virtex UltraScale+ XCVU19P device [8]. Synthesis and Place & Route (PnR) results show a highly optimized resource utilization profile (see Fig. 2(b)). The total Look-Up Table (LUT) utilization of the proposed FEC is 37%, and the Block RAM (BRAM) utilization reaches 68%. As illustrated in Fig. 2(a), the decoder consumes the majority of resources (88% of used LUTs and 73% of BRAMs in FEC total usage), while the encoder maintains a significantly

(a) Resource utilization breakdown of the OSA-SBC-FEC Module

Encoder	LUT(K)		BRAM	
	168.2	%	379	%
BCH encoder	62.624	4	-	-
Encoding Framer	50.118	4	313.5	22
Aggregator	55.397	4	65.5	5
LUT: 12 % BRAM: 27 %				
LUT: 1388.6K BRAM: 1424				

Decoder	LUT(k)		BRAM	
	1220.4	%	1045	%
BCH decoder(x4)	914.29	65	424	30
OSA decoder	200.18	14	196.5	14
Framer	66.483	5	321.5	23
Deframer	48.966	4	86.5	6
LUT: 88 % BRAM: 73 %				



(b) Total resource usage

(c) Floorplan & PnR result

Fig. 2: FPGA implementation results.

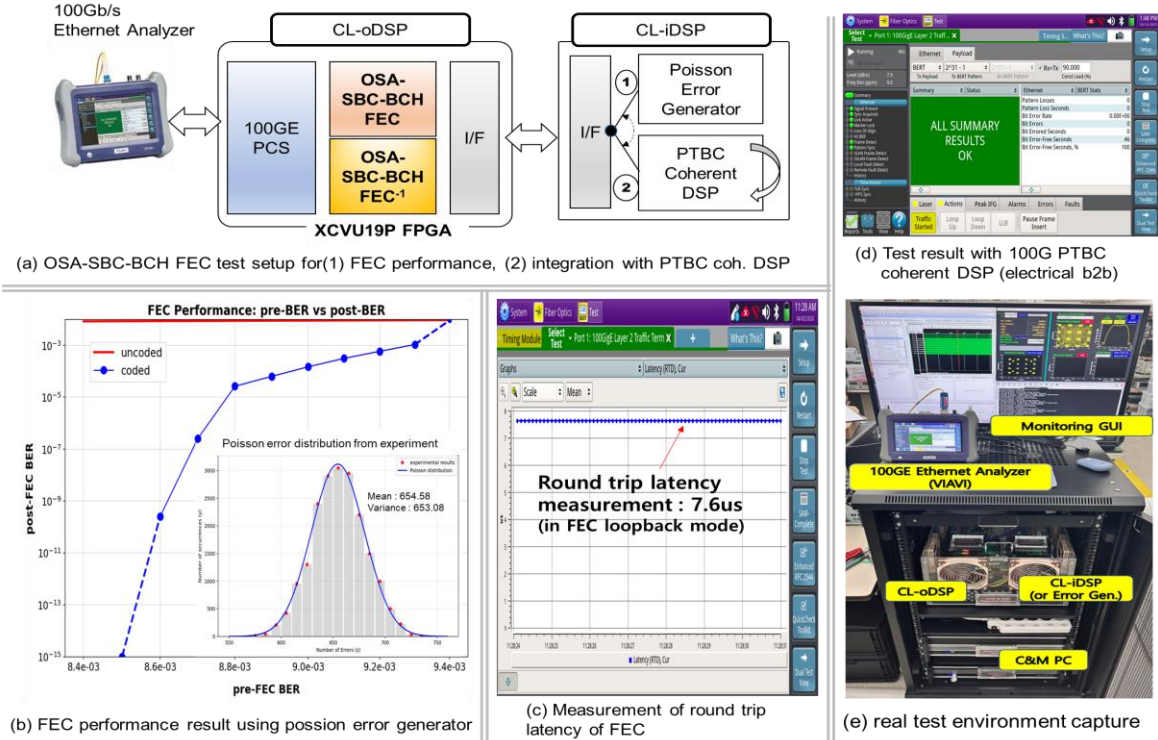


Fig. 3: Test environment and results of OSA-aided SBC-BCH FEC system.

lower profile. Within the decoder, the parallel BCH pipelines requires 65% of LUTs to sustain 100 Gbps throughput, while the OSA core efficiently uses only 14%. The high BRAM usage is heavily driven by the Framers and BCH decoders for buffering the 152×152 symmetric block matrices.

Furthermore, to overcome severe timing challenges from high-density parallel routing, the floorplan was meticulously optimized during the PnR (Place & Route) stage. As shown in Fig. 2(c), major blocks (ENC, OSA, BCH0–BCH3, and PCS) were strategically partitioned into dedicated physical regions. This minimizes long-path delays, ensuring reliable 100Gbps real-time operation.

B. System-Level Verification

To validate the real-time performance in a practical environment, an integrated testbed was constructed in a rack-mountable enclosure (see Fig. 3(a), (e)). The testbed comprises a VIAVI 100GbE tester [9], the proposed FEC FPGA (CL-oDSP), and a separate PTBC-based Single-Polarization Coherent DSP FPGA (CL-iDSP, also serving as an error generator for FEC performance test).

The verification was conducted in three distinct phases to confirm the performance, latency, and integration test. First, the FEC performance was analyzed using the hardware Poisson error generator to simulate various channel conditions. The Poisson error generator is used to emulate burst-like error behavior observed in practical optical links, providing a

more conservative evaluation compared to AWGN-based models [10]. The architecture achieves a post-FEC BER of 2.5×10^{-10} at a pre-FEC BER of 8.6×10^{-3} (see Fig. 3(b)). The inset in the graph shows the experimental Poisson error distribution, which closely aligns with the theoretical model, ensuring the statistical validity of the high-gain FEC performance. Second, the measured round-trip delay in the FEC loopback configuration (including the PCS and interface logics) is $7.6 \mu\text{s}$, which is lower than that of concatenated BCH SuperFEC, whose latency is $11.8 \mu\text{s}$ at 430 MHz [2] (see Fig. 3(c)). The proposed implementation achieves lower latency than [2], despite operating at a lower clock frequency. Finally, an integration test was performed with the PTBC-based coherent DSP to verify the back-to-back (B2B) operation. The 100GbE analyzer confirmed stable traffic transmission with 100% bit error-free operations.

Conclusion

This work demonstrates the first real-time FPGA implementation of OSA-assisted decoding for high-gain HD-FEC in 100G coherent optical access systems. The proposed architecture efficiently maps the entire FEC with 37% LUT and 68% BRAM utilization. System-level verification confirms robust error correction, achieving error-free operation with a low round-trip delay of $7.6 \mu\text{s}$, which is lower than high-gain BCH-based FEC implementations. These results validate the feasibility of low-latency, high-reliability FEC for next-generation optical access networks.

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